Workshop — Setup instructions for macOS

Before the workshop, set up the following on your laptop. This sheet is for **macOS**.

1. **Workshop folder:** download it from GitHub (Part 1).
2. **Vensim PLE:** everyone installs this (Part 2).
3. **For the hands-on exercises, choose one option**, whichever you find easier:
   - **Google Colab (recommended):** runs in your web browser. Nothing to install, but you need internet throughout (Part 3).
   - **Python + JupyterLab on your Mac:** a one-time installation, for those familiar with Python (Part 4). VS Code or PyCharm users can open the notebooks there instead.
4. **AnyLogic PLE** (Part 5).

Part 1 — Download the workshop folder (everyone)

1. Open github.com/IITB-TUD-DMDU/IITB-TUD-DMDU-Workshop in your web browser.
2. Click the green **Code** button, then **Download ZIP**.
3. In Finder, choose **Go → Home**, then **File → New Folder**, and name it `DMDU`.
4. In your Downloads folder, double-click the ZIP file (Safari may already have unzipped it). Drag the folder `IITB-TUD-DMDU-Workshop-main` into `DMDU`.

What you should see: the folder `DMDU/IITB-TUD-DMDU-Workshop-main` in your home folder. It contains the readings (`reading-pdfs`) and `requirements.txt`. The exercise notebooks, models and data will be added before the workshop; download the ZIP again when the organisers announce it.

Avoid iCloud Drive, Desktop or Documents if they sync to iCloud, and folder names with spaces; they are a common cause of errors.

Part 2 — Install Vensim PLE (everyone)

Vensim PLE is free for educational use. You use it to open and look at the workshop models.

1. Go to the Vensim free download page.
2. Choose **Vensim PLE** and **macOS**, and enter your email address.

3. Open the email from Vensim and click the download link. Check your spam folder if nothing arrives within 15 minutes.
4. Open the downloaded file and follow the installer, entering your Mac password when asked. No registration code is needed (Vensim installation help).
5. If macOS says Vensim cannot be opened, go to **System Settings → Privacy & Security** and click **Open Anyway**.
6. Start Vensim PLE and use **File → Open Model…** to open a `.mdl` file from the workshop folder.

What you should see: the model diagram opens. If you change a model, save it under a new name as an `.mdl` file, because the Python notebooks read `.mdl` files.

Part 3 — Google Colab (Recommended)

1. Go to colab.research.google.com and sign in with a Google account.
2. Choose **File → Upload notebook** and select the exercise notebook (`.ipynb`) from your workshop folder.
3. **Upload the model and data files:** click the folder icon on the left (Files panel), then the upload icon, and select the `.mdl` and data files the notebook uses.
4. Click **+ Code** to add a cell at the top, paste this line and press Shift+Enter (takes 1–2 minutes). If Colab shows a **Restart session** button, click it.

```
%pip install -q ema_workbench pysd SALib
```

5. Run the remaining cells from top to bottom with Shift+Enter.
6. Choose **File → Save a copy in Drive** to keep your work.

Good to know: Colab deletes uploaded files after about 90 minutes of inactivity or when you close the tab. Your notebook in Drive is kept; repeat steps 3–4 when you come back.

Part 4 — Python and JupyterLab on your Mac

If you are already familiar with Python and Jupyter, you are welcome to use your own setup, such as JupyterLab, VS Code or PyCharm. Install the packages listed in `requirements.txt` in your environment and open the workshop notebooks there. If anything does not work, switch to Google Colab (Part 3).

Part 5 — AnyLogic PLE

AnyLogic Personal Learning Edition is free for classroom instruction and personal learning (AnyLogic editions).

1. Go to the AnyLogic PLE download page.
2. Fill in the required fields of the form, choose the macOS version that matches your Mac (Apple Silicon (M-series) or Intel; check under Apple menu → About This Mac) and click **Download**.
3. Open the downloaded file and drag AnyLogic into Applications. If it will not open, go to **System Settings → Privacy & Security** and click **Open Anyway**.
4. Start AnyLogic. If it asks for activation, request the free Personal Learning Edition key with your email address and enter the key you receive.

What you should see: AnyLogic opens on its Welcome page.